Question 1 **1 mark** 101

Youyi wants to buy a new cell phone from a company that offers 33 different types of cell phones. Each cell phone is available in 5 different colours. Determine the total number of available options.

Question 2 2 marks 102

Determine the central angle subtended by an arc length of 30 cm on a circle with a diameter of 20.6 cm. Express your answer in degrees.

Question 3 **3 marks** 103

A choir has 14 sopranos. Karin and Ellen are two of them. Determine the number of possible ensembles of 3 sopranos that could be formed if either Karin or Ellen, or both, must be included.

Question 4 3 marks 104

Loïc invests \$5000 at an annual interest rate of 6.75% compounded semi-annually. The value of his investment grows according to the following formula:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

where A is the value of the investment after t years

P is the amount of the initial deposit

r is the annual interest rate (as a decimal)

n is the number of compounding periods per year

t is the time in years

Determine, algebraically, the length of time it will take Loïc's investment to grow to \$9500.

Question 5 — 4 marks 105

Solve, algebraically, over the interval $[0, 2\pi]$.

$$\csc^2\theta - \csc\theta - 6 = 0$$

Note: A calculator is not required for the remaining test questions.

Question 6 **2 marks** 106

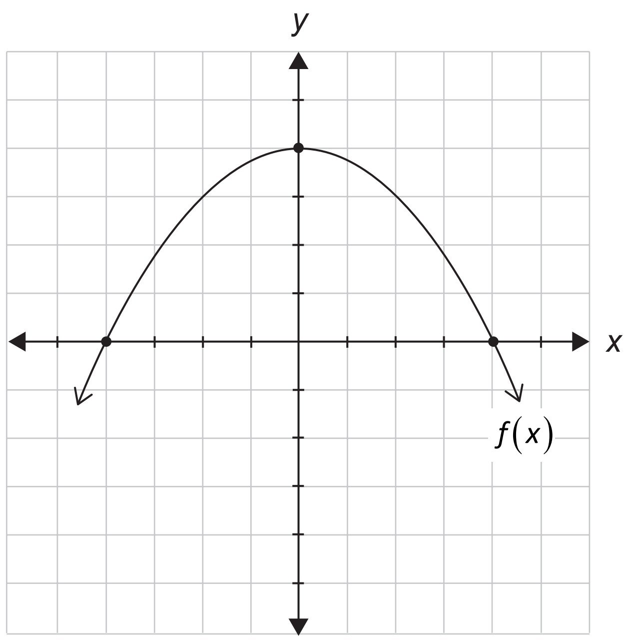
Determine how many 4-digit numbers can be made using the digits 0, 1, 2, 3, 4, and 5, if the third digit must be 3, and repetition is not allowed.

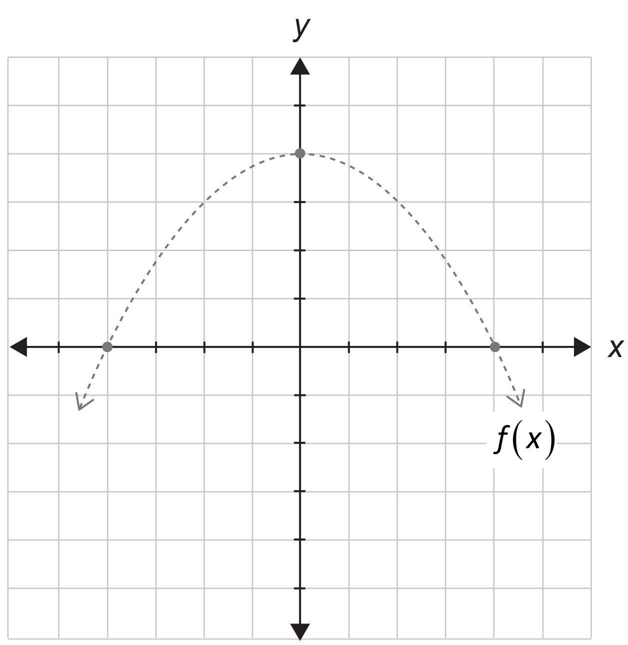
Question 7 3 marks 107

Describe the transformations required to obtain the graph of the function $y = -4f\left(\frac{1}{3}x\right)$ from the graph of $y = f(x)$.

Question 8 **2 marks** 108

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.





The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 9 **1 mark** 109

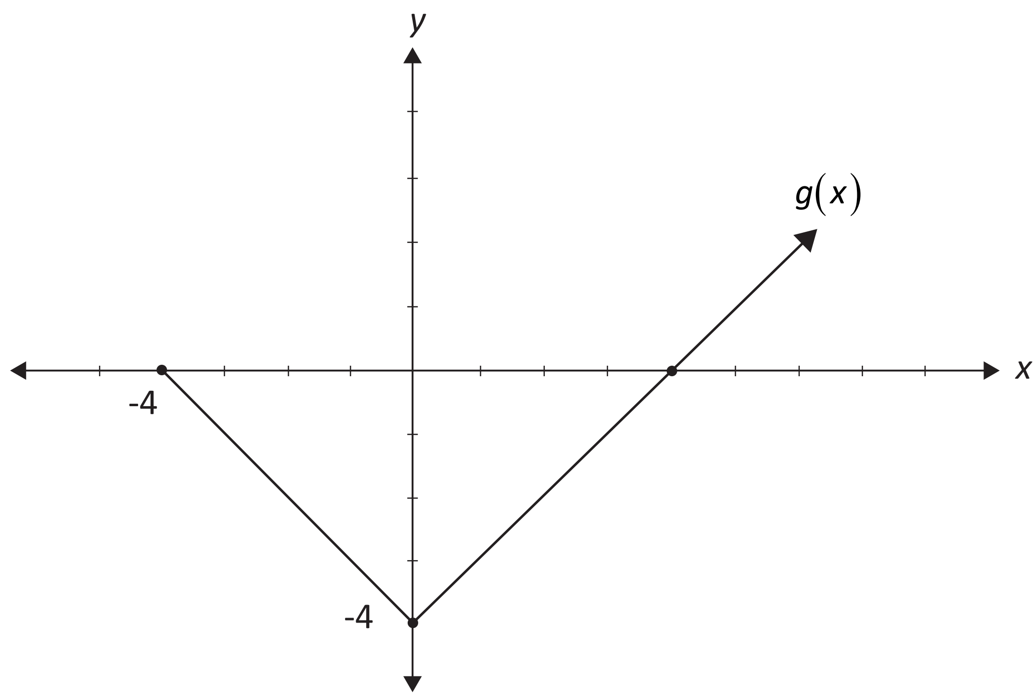
Describe how to determine if $(x-a)$ is a factor of the polynomial function, $p(x)$.

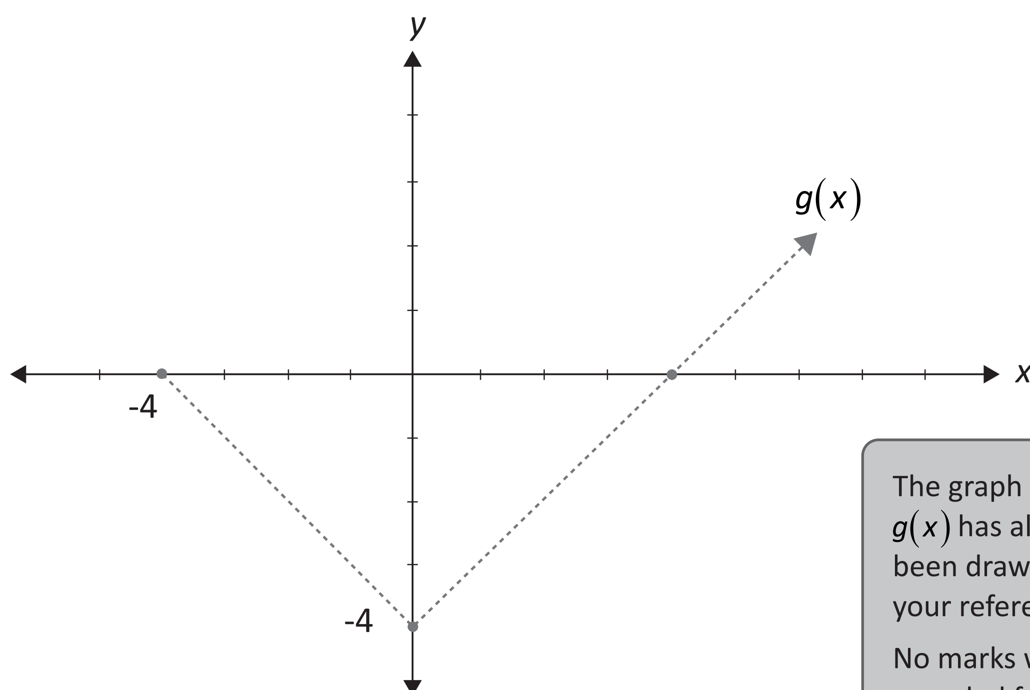
Question 10 **3 marks** 110

Express $\frac{1}{2}\log x - 2\log y + \log z$ as a single logarithm.

Question 11 **3 marks** 111

Given the graph of $g(x)$, sketch the graph of $f(x)=\frac{1}{g(x)}$.





The graph of $g(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $g(x)$.

Question 12 a) 1 mark b) 3 marks 112 113

Given the identity $\frac{1-\tan^2\theta}{\sec^2\theta}=\cos 2\theta$,

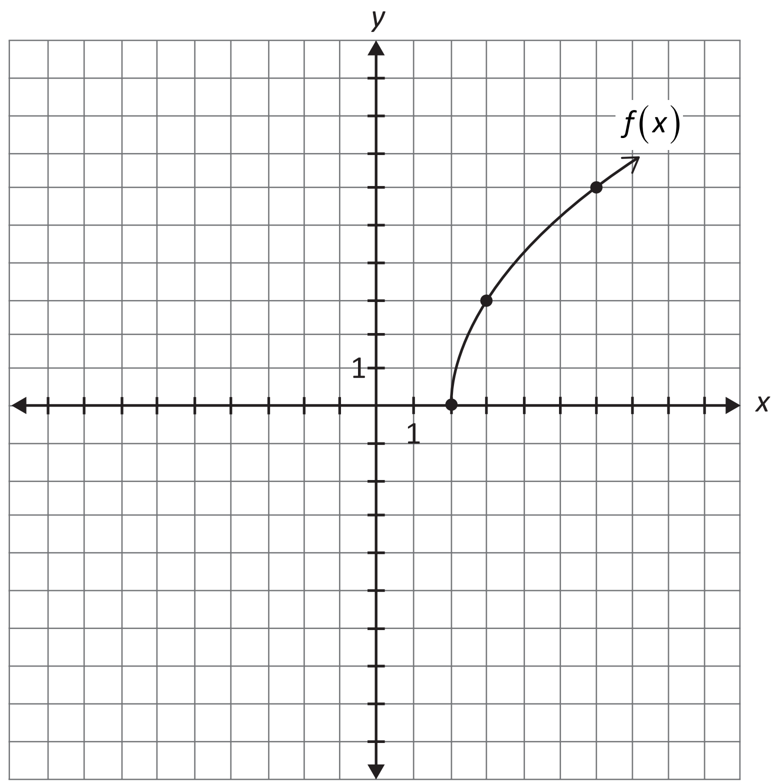
a) state a non-permissible value of θ.

b) prove the identity for all permissible values of θ.

Left-Hand Side	Right-Hand Side

Question 13 **2 marks**

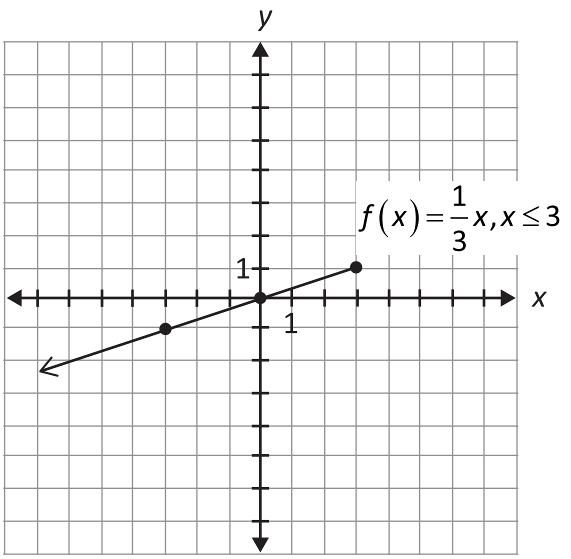
Determine an equation of the radical function represented by the graph of $f(x)$.

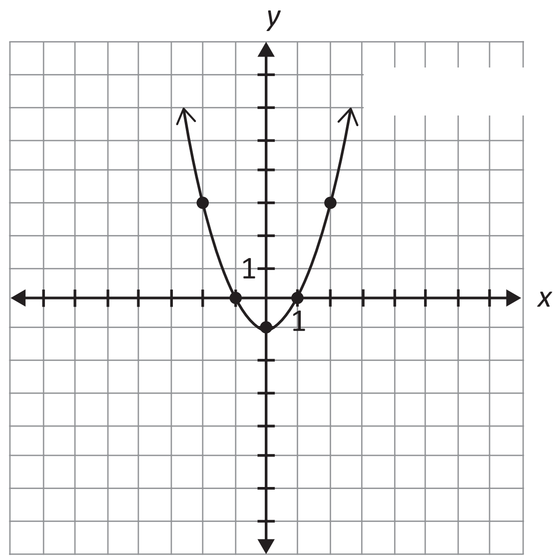


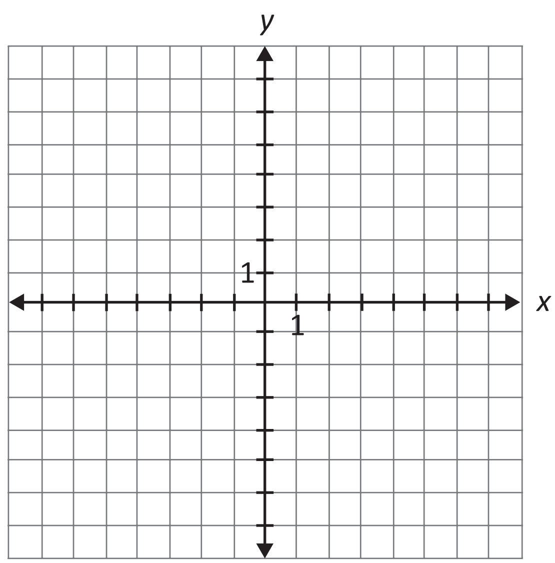
$f(x) =$ ______________________________

Question 14 **2 marks** 115

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(f(x))$.



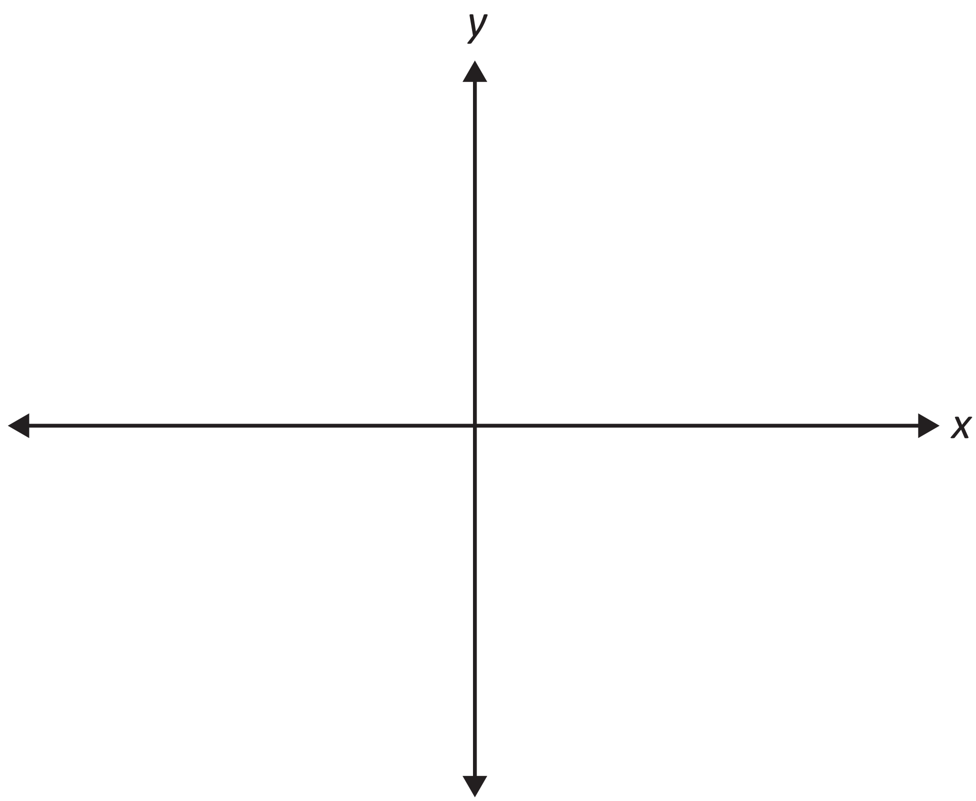




Question 15 **3 marks** 116

Sketch a graph of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 3
- $p(x)$ has a zero at – 5 with a multiplicity of 2
- $p(x)$ has a zero at 1
- $p(x)$ has a leading coefficient of – 2



Question 16 **2 marks** 117

Sketch the graph of $f(x)=\dfrac{x^2+x}{x}$.